

BIRJAND, Iran, Islamic Republic of

WMO: 408090

Lat: 32.8907N

Lon: 59.2839E

Elev: 1491

StdP: 84.65

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-6.0	-24.2	0.5	5.1	-20.8	0.7	7.0	10.8	8.6	9.4	8.7	1.6	120	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	38.2	16.2	37.0	15.7	35.8	15.3	18.3	35.5	17.3	34.3	16.3	32.9	4.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	9.9	22.8	10.0	9.2	23.1	8.9	8.5	22.4	58.0	35.8	54.5	34.2	51.4	32.9	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.8	7.7	-12.6	40.5	2.9	1.1	-14.7	41.3	-16.4	42.0	-18.0	42.6	-20.1	43.4		
			-13.8	18.8	2.6	2.0	-15.6	20.3	-17.1	21.4	-18.6	22.6	-20.5	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	4.3	6.8	12.0	17.4	22.7	26.8	28.2	25.9	22.4	17.3	10.5	6.1
	DBStd	8.95	4.21	4.48	4.23	3.88	3.31	2.79	2.35	2.48	2.87	3.50	3.95	3.78
	HDD10.0	487	181	106	28	2	0	0	0	0	0	0	41	129
	HDD18.3	1699	434	322	199	61	5	0	0	0	4	61	234	379
	CDD10.0	2956	4	17	91	224	393	504	565	493	371	227	57	9
	CDD18.3	1126	0	0	3	33	139	254	307	235	125	29	1	0
	CDH23.3	15761	0	3	63	454	1858	3522	4177	3154	1909	595	23	1
	CDH26.7	8242	0	0	13	112	858	2015	2427	1714	929	172	1	0
Wind	WSAvg	3.1	2.4	2.7	3.1	3.1	3.4	3.9	4.5	3.8	2.9	2.5	2.2	2.3
Precipitation	PrecAvg	172	35	37	40	19	7	1	0	1	2	3	8	21
	PrecMax	276	74	71	143	54	33	8	1	12	35	22	28	78
	PrecMin	72	9	4	4	0	0	0	0	0	0	0	0	0
	PrecStd	53	16	22	30	16	8	2	0	2	7	5	8	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	23.1	28.7	31.8	36.2	40.0	40.2	39.4	36.0	32.2	26.0	21.7
		MCWB	7.2	8.9	11.3	12.6	14.9	16.6	17.5	17.5	14.2	13.1	11.7	8.7
	2%	DB	16.8	19.4	24.9	29.6	34.4	38.1	38.8	36.9	34.4	30.2	23.4	18.2
		MCWB	6.6	8.1	10.2	12.8	14.5	16.1	16.6	15.8	13.8	12.9	10.0	7.3
	5%	DB	14.8	17.1	22.7	27.9	33.0	36.8	37.1	35.2	33.2	28.6	21.4	16.2
		MCWB	5.8	7.2	9.4	12.1	14.1	15.5	16.1	15.1	13.5	12.2	9.2	6.3
	10%	DB	12.5	15.1	20.8	26.1	31.2	35.1	35.8	33.8	31.9	26.8	19.4	14.2
		MCWB	4.8	6.3	8.9	11.3	13.7	14.9	15.7	14.5	13.3	11.4	8.3	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.8	12.9	14.6	16.6	18.5	20.0	19.0	16.9	15.0	13.0	10.0
		MCDB	13.9	17.8	20.9	27.3	32.9	37.6	36.5	35.7	32.9	28.8	22.7	16.8
	2%	WB	8.0	9.5	11.7	13.8	15.7	17.4	18.7	17.5	15.4	13.8	11.3	8.6
		MCDB	13.9	16.2	20.4	26.2	31.5	35.8	35.6	34.1	31.5	27.7	19.7	15.6
	5%	WB	6.9	8.6	10.7	13.0	14.9	16.4	17.7	16.4	14.5	12.8	10.1	7.5
		MCDB	13.4	15.0	19.4	24.6	30.2	34.1	34.6	32.8	30.5	26.6	19.2	14.3
	10%	WB	5.5	7.5	9.8	12.2	14.2	15.5	16.7	15.5	13.6	11.9	9.1	6.5
		MCDB	11.2	13.7	18.5	23.2	29.0	32.8	33.1	31.6	29.8	25.4	18.1	12.9
Mean Daily Temperature Range	5% DB	MDBR	13.6	13.1	13.6	14.0	14.8	15.1	14.2	15.7	17.8	17.4	15.6	14.9
		MCDBR	15.8	15.3	15.8	15.8	16.3	16.4	16.1	17.0	18.7	17.9	17.2	17.1
		MCWBR	8.8	7.7	6.9	6.5	6.3	6.2	5.9	6.7	8.1	8.3	9.0	9.6
	5% WB	MCDBR	13.1	12.0	12.7	13.8	14.7	15.3	14.4	15.1	17.1	16.9	14.8	13.9
		MCWBR	7.7	6.5	6.1	5.8	6.0	6.4	6.2	6.5	8.0	8.3	8.0	8.3
Clear-Sky Solar Irradiance	taub	0.315	0.365	0.441	0.506	0.518	0.479	0.452	0.413	0.375	0.384	0.348	0.306	
	taud	2.341	2.157	1.926	1.802	1.788	1.884	1.966	2.073	2.161	2.117	2.244	2.385	
	Ebn at Noon	897	876	830	793	787	815	836	865	885	841	845	888	
	Edh at Noon	100	133	182	215	220	200	183	162	141	137	109	91	
All-Sky Solar Radiation	RadAvg	3.27	4.07	5.29	6.32	7.24	8.29	8.17	7.67	6.56	5.09	3.74	3.15	
	RadStd	0.33	0.34	0.31	0.37	0.35	0.26	0.13	0.19	0.15	0.18	0.22	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.42	N/A	-3.50	+0.71	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.42	N/A	-3.50	+0.71	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page